# Engineering Professional Registration Manual Annex A

## Routes to Registration Guide

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# Introduction

**Welcome to Professional Engineering at Jaguar Land Rover!** This annex to the Professional Registration Manual is the guide for all Technicians and Engineers to select the most appropriate route, category and Institution for professional registration at Jaguar Land Rover (JLR).

## Competence

To achieve registration each individual's competence and commitment is independently and thoroughly assessed by their peers from the Engineering Council UK ($EC^{UK}$) licensed Professional Engineering Institution (PEI) they have chosen to apply to. These assessments are carried out against the standards laid down in the UK Standard for Professional Engineering Competence (UK Spec) published by the $EC^{UK}$. The 5 Competency areas are:

A. Knowledge and understanding
B. Design and development of processes, systems, services and products
C. Responsibility, management or leadership
D. Communication and inter-personal skills
E. Professional commitment

For each of the above competence areas there are levels of competence which are described below in general terms. To be fairly sure of being competent enough to gain registration it is widely accepted that for which ever category the individual is applying for they should be demonstrating level 3 across the 5 areas. More detail and specifics are available through the various PEI websites and representatives.

| Level | Competence Descriptors |
|---|---|
| 1<br>Aware | Performs the activity with significant supervision and guidance<br>Performs basic routine and predictable tasks<br>Little or no individual responsibility |
| 2<br>Familiar | Performs the activity in a range of contexts<br>Supervision only necessary in more complex circumstances<br>Some individual responsibility or autonomy |
| 3<br>Skilled | Performs activity in some complex and non-routine contexts<br>Significant responsibility and autonomy<br>Can oversee others' work |
| 4<br>Expert | Performs the activity in a wide range of complex and non-routine contexts<br>Substantial personal autonomy<br>Can develop others in the activity |

## Time

The time taken to become competent will vary between individuals, based on their ability, work load and external factors such as family and sporting commitments. It is entirely possible for an apprentice to be able to demonstrate reasonable ICTTech, or EngTech, competence levels before they complete the apprenticeship and this is the point of apprenticeships. Some graduates can easily demonstrate CEng competence with 3 years of graduation or IEng within 2 years, but for some it might take 6 or more years.

# Applicability and Target Groups

In line with the holistic and inclusive intent at JLR the following groups of technical professionals are the 'target audience'. It should be noted that each of the 4 categories is different and not better or lesser than the others. Some types and routes into engineer and the possible registration categories achievable are indicated below:

- Technicians with level 3 Technical Qualification – Engineering Technician (EngTech) or Information and Communication Technology Technician (ICTTech)
- Undergraduate Engineers with Level 3, Foundation Degree or 2 years of degree – EngTech
- Engineers with Bachelors Degree – Incorporated Engineer (IEng)
- Engineers with Masters Degree – Chartered Engineer (CEng)
- Engineers with few, or no, qualifications depending on experience to any category
- Professionally Registered Engineers with ICTTech, EngTech, IEng to the next category
- Professionally Registered Engineers from Member to Fellow

# EC$^{UK}$ Routes to Registration

There are many routes to registration and they do not necessarily have to involve formal qualifications, however qualifications are easier to verify than reports and sponsors comments. The whole point of professional registration is demonstration and independent verification of competence through what the applicant knows and their verified experience. Overleaf are flow charts adapted from the EC$^{UK}$ website that show the general academic requirements and available routes to registration for experienced and early careers pathways. Specific guidance should be sought from Early Careers Team, the Professional Engineer development Committee (PEDC) and PEI representatives if individuals are unsure of the appropriate route and category.

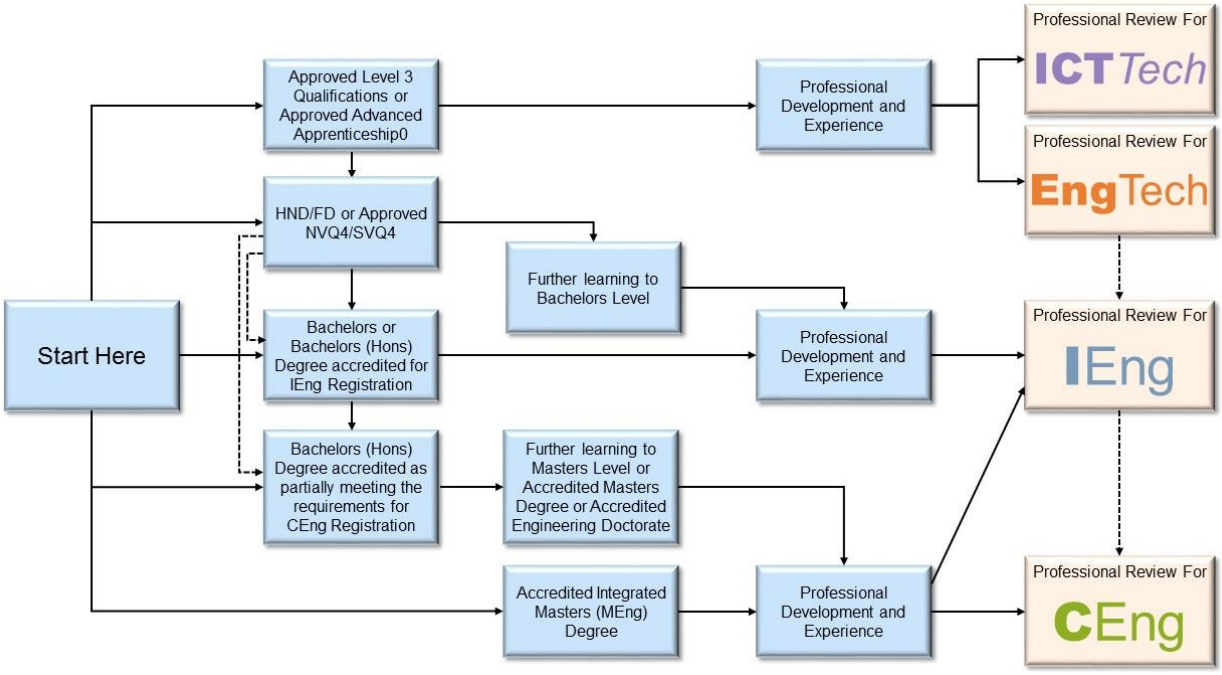


*Figure 1: EC$^{UK}$ Routes to Professional Registration*

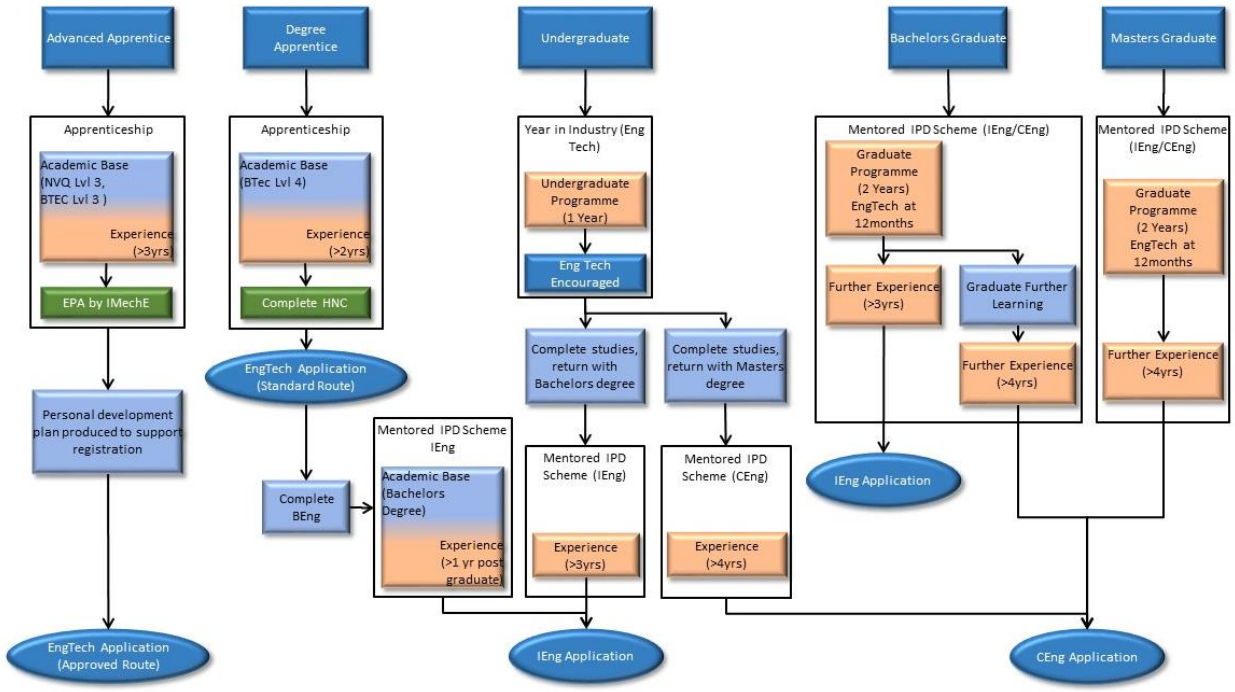


*Figure 2: Progression Flowcharts for Early Careers Talent Programmes*

# Scheme or Direct?

Developing Professionals on Early Careers Talent Programmes are encouraged to join an institution at a level of membership during their first 2 months via scheme awareness and induction sessions. Engineers recruited into a direct role should be encouraged by their managers to register on professional development sessions where the benefits of registration to both the individual and to JLR are promoted. These are coordinated and run by the HR Competence Developer Professional (HRCDP) and individuals will be guided to either Scheme or Direct application based on qualifications, knowledge and experience. An outline of appropriate routes for apprentice (Technicians and Degree), Undergraduates and Graduates follows in this document. Those not on schemes are encouraged to explore which of these routes fits their experience and qualifications. Technicians and Engineers do not need to have been on a scheme to develop and apply for registration.

## SuccessFactors – Mentoring Programme

All Technicians and Engineers wishing to become professionally registered must register on the SuccessFactors (SF) Mentoring Programme to assist with finding a suitable mentor, or supporter, for their development and application. Instructions for the SF system can be found at Annex C to the manual. The mentoring programme will enable tracking of JLR's professional Engineers and Technicians which is vital for the continued accreditation of apprenticeship and graduate schemes. Registered Engineers on the system will allow easier verification of entitlement to fee reimbursement and allocation of mentors / supporters for future applicants.

# Licensed PEIs

There are 34 PEIs licensed by EC$^{UK}$ to award some or all of the professional titles mentioned previously. Some of these PEIs are large organisations and very well subscribed and support at JLR, but others are much smaller and not commonly subscribed to by JLR engineers and technicians. The complete list is shown below with links to the main homepage. Engineers and technicians are encouraged to seek advice if they are not sure which institution is most suitable, but IET and IMechE are the most heavily subscribed and supported across the organisation.

Institution of Mechanical Engineers (IMechE)

Institution of Engineering and Technology (IET)

BCS, The Chartered Institute for IT

British Institute of Non-Destructive Testing (BINDT)

Chartered Institution of Building Services Engineers (CIBSE)

Chartered Institution of Highways & Transportation (CIHT)

Chartered Institute of Plumbing and Heating Engineering (CIPHE)

Chartered Institution of Water and Environmental Management (CIWEM)

Energy Institute (EI)

Institution of Agricultural Engineers (IAgrE)

Institution of Civil Engineers (ICE)

Institution of Chemical Engineers (IChemE)

Institute of Cast Metals Engineers (ICME) - licence suspended on 20 April

Institution of Engineering Designers (IED)

Institution of Fire Engineers (IFE)

Institution of Gas Engineers and Managers (IGEM)

Institute of Highway Engineers (IHE)

Institute of Healthcare Engineering and Estate Management (IHEEM)

Institution of Lighting Professionals (ILP)

Institute of Marine Engineering, Science & Technology (IMarEST)

Institute of Measurement and Control (InstMC)

Institution of Royal Engineers (InstRE)

Institute of Acoustics (IOA)

Institute of Materials, Minerals and Mining (IOM3)

Institute of Physics (IOP)

Institute of Physics and Engineering in Medicine (IPEM)

Institution of Railway Signal Engineers (IRSE)

Institution of Structural Engineers (IStructE)

Institute of Water

Nuclear Institute (NI)

Royal Aeronautical Society (RAeS)

Royal Institution of Naval Architects (RINA)

Society of Environmental Engineers (SEE)

The Society of Operations Engineers (SOE)

The Welding Institute

# Apprentices

Apprentices for the purposes of this guide fall in to 2 main categories; Level 3 Technicians and Degree apprentices however both groups will naturally qualify and should be capable of demonstrating competence as an EngTech early on in their career. Guidance for each type is below:

### Competent Level 3 Technicians

All Level 3 apprentices on successful completion of End Point Assessment (EPA), which is aligned to the 5 competence areas laid down in the UKSpec, should be competent enough to complete a successful direct application. The JLR apprenticeship schemes are accredited by the IMechE and IET therefore an approved route application is all that is required in these cases. This application is completed online and requires the demonstration of continued commitment to development and ethics.

Technicians who qualified prior to the introduction of EPA will need to complete a standard route application as their competence across the 5 areas has not previously been assessed by a Professional Engineering Institution (PEI)

### Degree Apprentices

Apprentices currently on the Degree apprentice scheme may already have the underpinning knowledge in the form of a level 3 technical qualification such as a BTec. Once they can demonstrate competence as described in the EngTech section of the UKSpec apprentices are encouraged make a direct standard route application for EngTech with an appropriate PEI. This is can be a good stepping stone and experience for working towards IEng or CEng on completion of their degree.

On completion of the degree apprenticeship the newly qualified, but experienced apprentice is likely to be demonstrating the required levels of competence for a direct IEng Application, but at this stage will lack the masters level of demonstrable knowledge for a CEng Application.

*Note:* It is not recommended that Degree apprentices join one of JLR's accredited professional development schemes (MPDS or Carer Manager), as the additional work load writing quarterly reports would not be appropriate whilst still completing their degree.

# Undergraduates (Year Placement)

Undergraduates on the year placement scheme will all have completed at least 2 years of an engineering degree and may also have gained a level 3 technical qualification before going to university. Their technical knowledge will be more than adequate to meet the criteria for EngTech and after 11 months working on real projects they should be able to demonstrate competence on a standard route EngTech application with a suitable PEI. Applications for EngTech should be completed and submitted whilst still at JLR to enable support and sponsorship by registered engineers or appropriate managers who can verify the competence of the applicant.

Many engineering undergraduates will be on a university MPDS scheme and EngTech application does not preclude this development activity from continuing when back at university or after graduation. EngTech will enhance self-awareness of competence and enable full PEI membership with legally protected post-nominals to demonstrate verified competence.

# Graduates

When joining JLR on the Engineering and Manufacturing graduate schemes, graduates are strongly encouraged to work towards professional registration in IEng or CEng on the IMechE MPDS scheme or IET's Career Manager. Mentors for the schemes are registered on the SF Mentoring Programme and available to help develop graduates all graduates. This also applies to recently graduated engineers who have joined the company, but are not on the graduate scheme and indeed any person with a degree who wishes to become registered.

New Graduates should also be encourage to apply for EngTech by the end of their first year at JLR, as this will demonstrate some level of competence and provide valuable credibility across the business and with customers and suppliers.

There is no specific institute that the graduates must join, but the level of support for the IMechE and IET is much greater than for other institutions. Only those institutions licensed by $EC^{UK}$ are supported by the engineering professional registration manual. Systems such as; MPDS, Career Manager and MyCareerPath are all great tools for logging achievements and progress in readiness for final application for IEng or CEng. Career Manager is freely available to all those who are a member of the IET or have an IET online account. Guides for MPDS and Career manager are linked from the main manual in the resources section.

While mentors are expected to support graduates working towards registration, it is the responsibility of the DE to ensure they submit their quarterly and annual reports, meet up with their mentor regularly and continuously look to develop. The purpose of a mentored and monitored scheme is to drive development in a controlled and sustained manner, so registration and professional competence is achieved in a timely manner.

Graduates with 3 or more years relevant experience are best advised to assess their competence and consider direct application with guidance from an experienced registered engineer to mentor them through the application process and support / sponsor the final application.

# Accredited Schemes

JLR currently operates IPD schemes accredited by the IMechE and IET. The delivered content of these schemes at JLR is common between institutions; however the processes for planning, recording and reviewing development, and the systems provided by the engineering institutions to facilitate this process are specific to the Institution. Even if DEs are not on the scheme or intending on applying to the IMechE or IET they can still benefit from the processes and procedures put in place to support professional development and registration.

### IMechE Monitored Professional Development Scheme (MPDS)

Developing IMechE members working towards IEng or CEng are encouraged to use the IMechE MPDS system for planning, logging and reporting on their development and will follow all requirements of this system/process with regards to planning, recording and reviewing evidence. Engineers already on an MPDS scheme with their university will transfer to the JLR scheme (PDS574) on graduation. The annual fee for MPDS is paid by invoice direct from JLR to the IMechE whilst DEs are reasonably up to date with quarterly reports as laid down in the main manual document.

### IET Accredited Professional Development Scheme

All Developing IET members are encouraged to use the 'Career Manager' online system for planning, recording and reporting on their development. A JLR accredited scheme (scheme number 2000) is available for engineers working towards IEng or CEng to enrol on. Acceptance onto the scheme is carried out by the Scheme Administrators. IET members will be expected to update their Career Manager plans, records and review them with their mentor on a quarterly basis.

## Further Learning

In order to achieve any registered technician or engineer status in any of the categories it is necessary to demonstrate an appropriate level of knowledge. There are several ways in which knowledge gaps can be bridged or demonstrated and each institute has its own specific guidelines and methods. Firstly EC$^{UK}$ and the PEIs maintain lists of accredited and approved qualifications. Guidance should be sought from PEDC and the Professional Development Team if your qualification does not appear on the approved list or you do not have a qualification at the level indicated for the desired category of registration.

In some case it might be appropriate for the individual to carry out some formal further learning to bridge the knowledge gap. Further learning is available for engineers in various forms and can in the majority of cases be provided through formal learning modules, on SF Learning, up to and including the accredited TAS (Engineering and Scientific deepening) modules for Masters level knowledge. TAS modules will require a business case by the individual to justify the cost of the modules to their local management team. Given the TAS modules are to improve technical ability and competence in the role making the case for the learning to be funded and time allowed to complete should not be too difficult and will assist the developing engineer in demonstrating competence C. TAS Modules form part of an IMechE accredited Further Learning Scheme details of the full process can be found in Annex D to the Engineering Professional Registration Manual.

## Professional Application and Review Workshops

If you and a supporter think you are ready to apply for registration, no matter what route you have taken to get to this point, help is available to guide you through the process. Workshops will be arranged by the HRCDP and made bookable on SF to assist individuals to prepare for the professional application and review process. Indicative content of these sessions is:

- Application process overview
- Assistance with getting started on the application including application writing advice
- Application document review
- Interview format information and mock interviews

For those applicants overseas support may be provided via video, telephone conference or email. Interviews overseas will need to be arranged by the applicant directly with the institution to which they are applying. Institutions may also provide support with interview preparations overseas.

## Final Application Processes

These sessions are the common processes for ICTTech, EngTech, IEng and CEng Applications. EngTech candidates are not normally interviewed and this guided process illustrated below can also be followed for applications for Career Learning Appraisal with the IMechE and Fellow applications for any PEI.

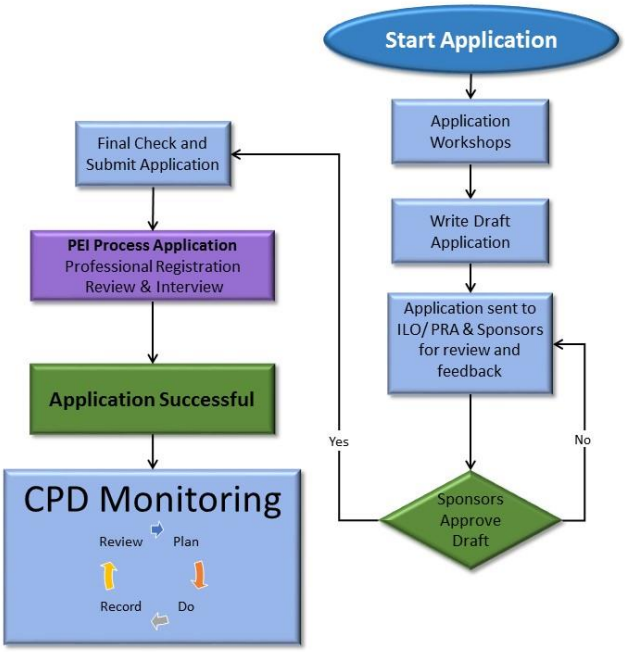


*Figure 4: Flowchart of final application process*

# Document History

| Issue | Date | Change history | By whom |
|---|---|---|---|
| **First Edition** | **6 Jun 2019** | Initial Issue of first edition based of various documents created by the graduate team and feedback from Mentors and Mentees | **Simon Wood swood14** |
| **First Edition Draft** | **19 Feb 2019** | Initial Draft Issue of first edition based of various documents created by the graduate team | **Simon Wood swood14** |